2 Introduction to Vibrations and Waves

2.1 VIBRATIONS

The general objective of this chapter is to give an introduction to vibrations and waves (see, e.g., Pain [1]). More specifically, this chapter also has the goal of recalling the basic mathematical apparatus necessary to read the book and to introduce the simple physical ideas and analogies that will be useful throughout the book. The model system used will be a simple oscillator, a mass connected to a spring, although a simple pendulum or any other similar system could have been used. For small displacements, it will be seen that the oscillations are sinusoidal at a single frequency, so-called simple harmonic motion.

From Figure 2.1, we easily see that the motion will be periodic. If the mass is displaced, initially there will be a restoring force due to the spring. For small displacements, Hooke's law applies, so that the restoring force is given by $F = -kx$. This is in fact the leading term in a Taylor's expansion of force in terms of displacement. Hooke's law is ubiquitous in mechanical problems of vibrations and waves. For example, it is this approximation that is used to define the elastic constants of crystals and that is also the basis of the theory of elasticity of solids. If Hooke's law is not obeyed, then things become much more complicated, mathematically and physically, and we enter the realm of nonlinear acoustics. Except where stated otherwise, we will always remain in the linear regime described by Hooke's law.

Hooke and Newton were great English scientists of the seventeenth century, and there was ill-concealed tension between them. It is thus somewhat ironic that the basic equation for the simple oscillator and the wave equation are both obtained by a happy combination of Hooke's law and Newton's equation of motion. For the mass-spring system, this can be written as

$$F = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} \tag{2.1}$$

or

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{k}{m}x = 0 \tag{2.2}$$

Physically, this equation provides the solution $x(t)$ for the displacement of mass. Once the mass is released at $t = 0$, it is pulled in the $-x$ direction by the spring, which is in turn compressed by the movement of the mass. At the moment of maximum compression, all the energy of the system is stored as potential energy in the spring. The mass is then repelled to the right by the spring, and at the instant where the spring extension is zero, the potential energy is also zero and all the energy of the system is now in the form of kinetic energy of the mass. If there is no dissipative force, the process will be periodic with exchange from kinetic to potential energy and vice versa and will continue ad infinitum. If there is dissipation, for example, friction with the supporting surface, the motion will be progressively damped and will finally come to a halt. Finally, it should be noted that this is a fixed, isolated vibrator that undergoes periodic motion. There is no wave propagated here: that aspect will be discussed in Section 2.2.

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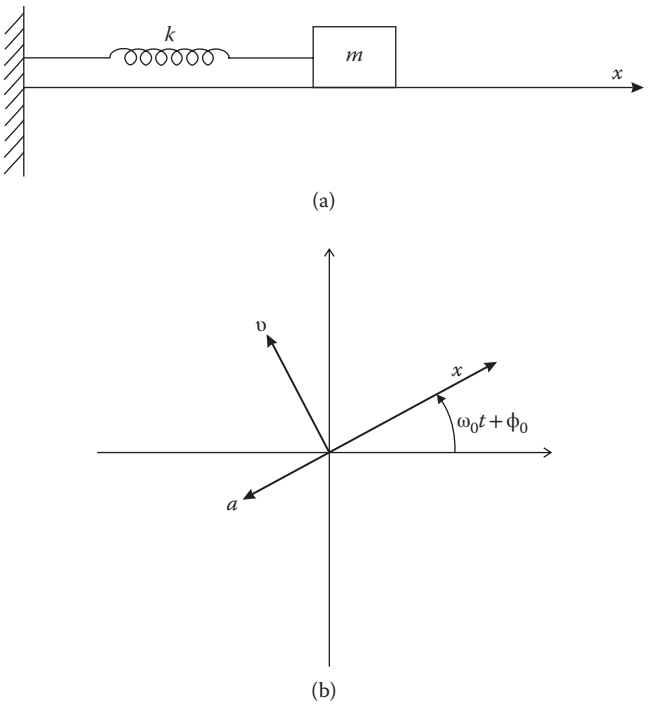

FIGURE 2.1 (a) Mass-spring oscillator. (b) Phasor diagram for simple harmonic motion.

Returning to Equation 2.1, this can be clearly identified as the harmonic equation, with harmonic solutions. Defining the angular frequency $\omega_0^2 = k/m$, these solutions are of the form

$$x = A_1 \cos \omega_0 t + A_2 \sin \omega_0 t \tag{2.3}$$

For this second-order homogeneous differential equation, the solution has two arbitrary constants to be determined by the initial conditions. Alternatively, the solution can be written as

$$x = A \sin(\omega_0 t + \phi_0) \tag{2.4}$$

where ϕ_0 is an initial phase angle. The frequency f and the period T are determined by

$$f_0 = \frac{\omega_0}{2\pi} \tag{2.5}$$

$$T = \frac{1}{f_0} \tag{2.6}$$

The subscript zero is used, as this is a simple undamped oscillator.

The complete solution can be found using the initial conditions. At $t = 0$, we define the initial displacement x_0 and the initial velocity v_0, from which we immediately find

$$A = \left(x_0^2 + \frac{v_0}{\omega_0}^{2} \right)^{\frac{1}{2}} \tag{2.7}$$

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$$\phi_0 = \tan^{-1} \frac{-v_0}{\omega_0 x_0} \tag{2.8}$$

which completely determines the displacement from Equation 2.4. The velocity v and acceleration a are immediately found as

$$v = v_m \cos(\omega_0 t + \phi_0) \tag{2.9}$$

and

$$a = -\omega_0 v_m \sin(\omega_0 t + \phi_0) \tag{2.10}$$

From these solutions, we can deduce that the displacement and velocity are in phase quadrature (displacement lags by $\pi/2$), and the displacement and acceleration are π out of phase. This type of analysis will be found to be important for waves.

2.1.1 Vibrational Energy

For a mechanical system, in general, the total energy U is the sum of the potential energy U_P and the kinetic energy U_K. These are readily calculated for our model system. U_P is determined by the work done to compress the spring:

$$U_P = \int_0^x kx \; dx = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \sin^2(\omega_0 t + \phi_0) \tag{2.11}$$

The kinetic energy is determined by the usual mechanical formula for a mass m:

$$U_K = \frac{1}{2}mv_m^2 = \frac{1}{2}mv_m^2 \cos^2(\omega_0 t + \phi_0)$$

Hence, the total energy is given by

$$U = U_P + U_K = \frac{1}{2}m\omega_0^2 A^2 \tag{2.12}$$

Alternatively, as could have been deduced from the discussion of energy exchange during a cycle, the total energy is simply equal to the maximum potential or kinetic energy:

$$U = \frac{1}{2}kA^2 = \frac{1}{2}mv_m^2 \tag{2.13}$$

2.1.2 Exponential Solutions: Phasors

The previous results for x, v, and a were obtained using the real trigonometric functions sine and cosine to represent the periodic variation with time. There is an alternative representation that is conceptually simple and mathematically more economic than the use of real trigonometric functions. This is the use of complex exponentials, which is almost universally employed in research papers. In the complex plane, it is well known that we can represent sine and cosine functions in the complex plane by using Euler's rule

$$e^{j\theta} = \cos\theta + j\sin\theta$$

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where $j = \sqrt{-1}$. Generally, j is used in engineering practice and i in mathematics and physics, but this is not universal. When they are not used as an index, the scalars i or j always represent $\sqrt{-1}$. We may use them interchangeably. In the complex plane, the x axis represents the "real" part and the y axis represents the "imaginary" part of a variable $z = x + iy = re^{i\theta}$. When a physical quantity is represented by a complex variable z, by convention its physically significant part is given by Re(z). This is pure convention; since the real and imaginary parts contain redundant information, the imaginary part could equally well have been chosen. The semantics have been chosen to reinforce the conventional choice.

Complex exponential notation is ideally suited for the representation of harmonic vibrations. Thus, instead of describing a physical displacement as $x = A \cos \omega t$, we can represent it by the quantity $x = Ae^{j\theta} = Ae^{j\omega t}$. The radius vector A is real and it rotates at constant angular velocity $\dot{\theta} = \omega$. Thus, the projection on the x axis, the real part, traces out the variation $x = A \cos \omega t$ with time. The polar representation is called the phasor representation (A is a "phasor"). Phasors are simple graphical way to represent vibrations, and they are particularly useful when several different vibrations are added and when one wishes to calculate the resultant. As before, two quantities must be given to specify a phasor, namely the amplitude (radius vector) and the phase (angle θ). Another analytical advantage of the use of complex numbers and phasors is that multiplication by j corresponds to an advance in phase by 90° (rotation from the real to the imaginary axis). Similarly, multiplication by $-j$ retards the phase by $\pi/2$. Thus, phase relationships can be deduced instantly from analytical formulae by identifying the imaginary terms and their sign.

2.1.3 Damped Oscillations

A simple undamped oscillator is, of course, an academic simplification. In the real world, there are always frictional and resistive effects that eventually damp out an oscillator's movement unless it is maintained by an external force. In this section, we examine the damping effects and then study the forced, damped oscillator in the subsequent section.

Most if not all damping mechanisms provide an opposing force that is proportional to the velocity or current. Frictional forces and the potential drop across a resistor are two common examples. The force can be written as

$$F = -R_m \frac{\mathrm{d}x}{\mathrm{d}t} \tag{2.14}$$

where the subscript m stands for mechanical, to distinguish R_m from an electrical resistance R. In a mass-spring system, R_m is often represented as a dashpot that slows the movement of the mass. The equation of motion can now be written as

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2} + \frac{R_m}{m}\frac{\mathrm{d}x}{\mathrm{d}t} + \omega_0^2 x = 0 \tag{2.15}$$

using a trial solution $x = Ae^{\gamma t}$

$$\gamma^2 + \frac{R_m}{m}\gamma + \omega_0^2 \quad x = 0$$

leading to a condition on γ

$$\gamma = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \tag{2.16}$$

where $\alpha = R_m/2m$.

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For typical mechanical systems of interest, the oscillation persists for at least several cycles so that $\alpha < \omega$ for this case. We then define a frequency $\omega_1^2 = \omega_0^2 - \alpha^2$ for the damped oscillator, so that finally

$$x = e^{-\alpha t}(A_1 e^{j\omega t} + A_2 e^{-j\omega t}) = Ae^{-\alpha t} e^{j(\omega_1 t + \phi)} \tag{2.17}$$

2.1.4 Forced Oscillations

In practice, virtually all oscillators are forced, either by external amplifiers or by feedback. Hence, the frequency response is of prime importance; depending on the application, the objective may be to excite the oscillator at a particular frequency or over a wide bandwidth. We start by establishing the system response at a single driving frequency and then extend these results to the response for an arbitrary frequency.

For an applied force $Fe^{j\omega t}$, the differential equation can be written as

$$\frac{d^2x}{dt^2} + \frac{R_m}{m}\frac{dx}{dt} + \omega_0^2 x = Fe^{j\omega t} \tag{2.18}$$

Physically, in the steady state, the system must respond at the applied frequency, so we look for solutions of the form $x = Ae^{j\omega t}$. Substitution in Equation 2.18 gives

$$x = \frac{1}{j\omega}\frac{Fe^{j\omega t}}{R_m + j\left(\omega m - \frac{k}{\omega}\right)} \tag{2.19}$$

and

$$v = \frac{dx}{dt} = \frac{Fe^{j\omega t}}{R_m + j\left(\omega m - \frac{k}{\omega}\right)} \tag{2.20}$$

Equation 2.20 has the form of Ohm's law for an electrical alternating current (AC) circuit. A formal analogy can be established by defining the mechanical impedance

$$Z_m = R_m + jX_m \tag{2.21}$$

where the mechanical reactance $X_m = \omega m - k/\omega$ follows from Equation 2.20. Analogous to Ohm's law, we then have impedance = force/velocity. This analogy is also valid for acoustic waves, and the concept of acoustic impedance will be used throughout this book.

Analogous to electrical circuits, the real and imaginary parts of the impedance can be represented by a vector diagram, corresponding to the complex plane, with phase angle $\tan\theta = [\omega m - k/\omega]/R_m$. The real values of displacement and velocity are given by

$$x = \frac{F}{\omega Z_m}\sin(\omega t - \theta) \tag{2.22}$$

$$v = \frac{F}{Z_m}\cos(\omega t - \theta) \tag{2.23}$$

Thus, the velocity lags the applied force by a phase angle θ. As in an AC circuit, this will affect the power transferred to the oscillator as the force and velocity are, in general, not in phase. The power transferred at time t is

$$P(t) = F(t)v(t) = \frac{F^2}{Z_m}\cos\omega t\cos(\omega t - \theta) \tag{2.24}$$

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Of more importance is the average power transferred over a cycle

$$P_0 = \langle P(t) \rangle = \frac{1}{T}\int_0^T P(t)\,\mathrm{d}t = \frac{F^2}{2Z_m}\cos\theta = \frac{F^2 R_m}{2Z_m^2} \tag{2.25}$$

The maximum power transferred occurs when the mechanical reactance vanishes ($\theta = 0$) and the impedance Z_m takes its minimum value R_m, which occurs at $\omega = \omega_0$. This is called the resonance frequency of the system. The power as a function of frequency is shown in Figure 2.2. An important parameter of the power curve $P_0(\omega)$ is the relative width of the curve around the resonance. Like the equivalent electrical system, the width is described by the Q or quality factor. There are various ways to define and describe the Q of the system and these are summarized as follows:

1. The Q can be defined as the resonance frequency divided by the bandwidth $BW \equiv$ frequency difference between the upper and lower frequencies for which the power has dropped to half its maximum value:

$$Q = \frac{\omega_0}{BW} \tag{2.26}$$

Hence high Q corresponds to a sharp resonance with a narrow bandwidth.

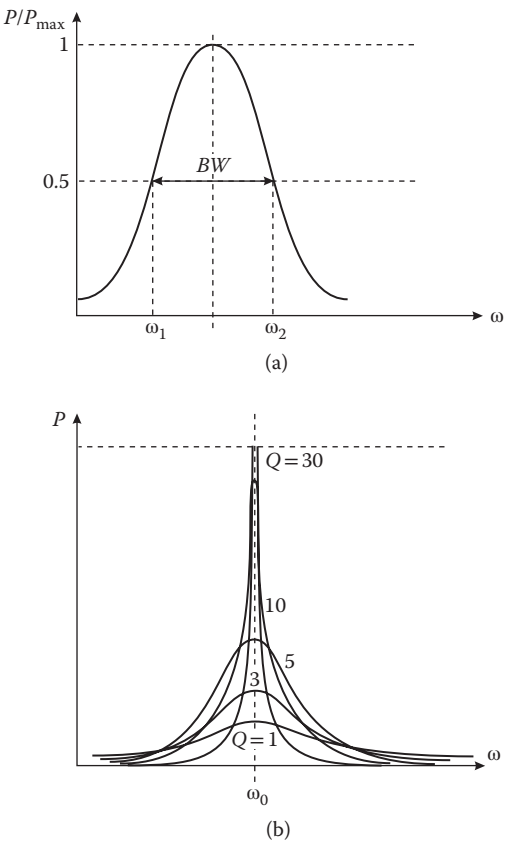

FIGURE 2.2 (a) Mean power input as a function of frequency to show the sharpness of the resonance curve. (b) Mean power absorbed by a forced oscillator as a function of frequency in units of $F^2\sqrt{2m\omega_0}$.

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2. The above form for Q can be rewritten in terms of mechanical constants. For the two half power points, $Z_m^2 = 2R_m^2$. Using $X_m = \omega m - k/\omega$, this gives

$$Q = \frac{\omega_0 m}{R_m} \tag{2.27}$$

   Thus, high Q corresponds to small R_m or low loss.
3. In terms of the decay time τ of the free oscillator, which is the time for the amplitude to fall to 1/e of its initial value, $\tau = 1/\alpha$ from Equation 2.17, $\alpha = R_m/2m$,

$$Q = \frac{1}{2}\omega_0\tau \tag{2.28}$$

   This means that a high Q oscillator when used as a free oscillator will "ring" for a long time, of the order of τ, before the amplitude falls to zero.
4. Finally, a formal definition of Q, equivalent to the above, is

$$Q = \frac{\text{stored energy}}{\text{total energy dissipated}} \tag{2.29}$$

   Again, a high Q oscillator is a low loss system.
5. Q can also be seen as an amplification factor. As R decreases, the displacement–frequency curve gets sharper and the amplitude at resonance A_0 increases significantly. Direct calculation of Q from the definition leads to

$$Q = A_0 \frac{k}{F_0} \tag{2.30}$$

F_0/k is the amplitude at asymptotically low frequencies, so Equation 2.30 means that the amplitude at resonance is a factor of Q greater than at low frequencies. This is the physical basis for the demonstrably high displacements attainable in mechanical systems at resonance. The same principle is routinely exploited in high Q electrical circuits, for example, in radio frequency receivers.

The full analogy between electrical and mechanical quantities is displayed in Table 2.1, together with a list of key formulae. Physically, by Lenz's law, inductance corresponds to the inertia (mass) of the system to change in current. The condenser stores the potential energy as does the compressed spring in the mechanical system. The resistance corresponds to the dissipated energy in both cases. Care must be taken in what quantities are held constant when comparing electrical circuits to mechanical configurations. For example, in Figure 2.3a, the source voltage is held constant and the same current flows through all elements in the electrical circuit. This clearly corresponds to the mechanical configuration shown in Figure 2.3b, where all elements have the same velocity and amplitude if the force is constant.

2.1.5 Phasors and Linear Superposition of Simple Harmonic Motion

A phasor has amplitude and orientation (phase angle) and as such is a vector. If two phasors have the same frequency, then they can be added vectorially. Graphically, they can be drawn head to tail to give a resultant phasor with components, as shown in Figure 2.4. For n such phasors, we have

$$\begin{aligned} A &= \left(\sum A_n \cos\phi_n\right)^2 + \left(\sum A_n \sin\phi_n\right)^2 \;^{\frac{1}{2}} \\ \tan\phi &= \frac{\sum A_n \sin\phi}{\sum A_n \cos\phi} \end{aligned} \tag{2.31}$$

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TABLE 2.1
Comparison of Equivalent Electrical and Mechanical Resonant Circuits

Electrical	Mechanical
Charge Q	Displacement x
Current I	Velocity v
Applied voltage V	Applied force F
Resistance R	Mechanical resistance R_m
Inductance L	Mass m
Capacitance C	Spring compliance $C = 1/k$
Impedance $Z = R + j(\omega L - 1/\omega C)$	Mechanical impedance $Z_m = R_m + j(\omega m - k/\omega)$
Differential Equation	
$L\frac{d_2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = V_0\, e^{j\omega t}$	$m\frac{d^2x}{dt^2} + R_m\frac{dx}{dt} + kx = F_0\, e^{j\omega t}$
Solution	
$Q = \frac{1}{j\omega}\frac{V}{Z}$	$x = \frac{1}{j\omega}\frac{F}{Z_m}$
Resonant Frequency	
$\omega_0 = \sqrt{1/LC}$	$\omega_0 = \sqrt{k/m}$
Energy	
$U_K = \frac{1}{2}LI^2$	$U_K = \frac{1}{2}mv^2$
$U_P = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$	$U_P = \frac{1}{2}kx^2$
Phase Angle	
$\phi = \tan^{-1}\frac{(\omega L - 1/\omega C)}{R}$	$\phi = \tan^{-1}\frac{(\omega m - k/\omega)}{R_m}$

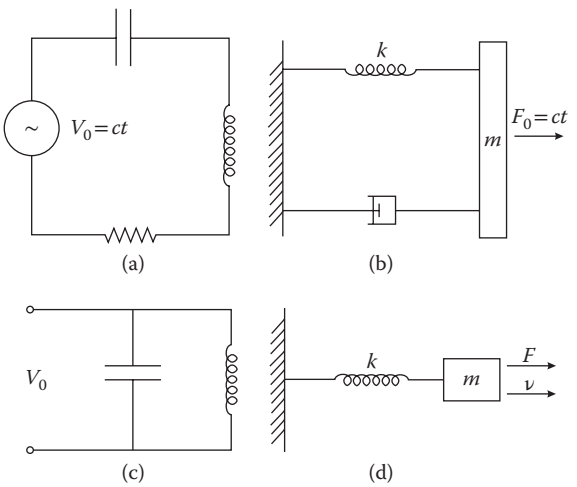

FIGURE 2.3 (a) Series electrical circuit and (b) its mechanical equivalent. (c) Parallel electrical circuit and (d) its mechanical equivalent.

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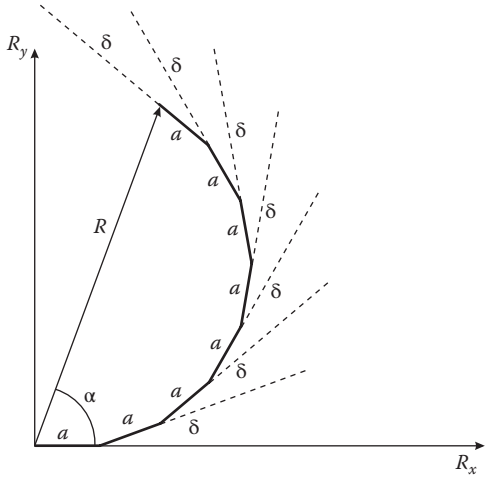

FIGURE 2.4 Addition of phasors of equal amplitude and phase difference.

For $n \to \infty$ and equal contribution for each constituent, the polygonal locus becomes an arc of a circle. In this way, interference and diffraction patterns in acoustics and optics can be constructed.

The above results are for superposition of vibrations at the same frequency. If the frequencies are different, the motion becomes complicated and aperiodic, even if there are only two components. In the case of two vibrations with frequencies very close together, "beats" can be observed at the difference frequency. The question will be taken up for the case of waves and the formation of wave packets later in this chapter.

2.1.6 Fourier Analysis

We now turn to what is in some respects the inverse problem to the addition of phasors presented in the last section. If we start with an arbitrary periodic function, Fourier showed that it can be represented as an infinite sum of sine and cosine (i.e., harmonic) terms. The subject, together with that of Fourier transforms for nonperiodic functions, has been treated in numerous texts and we only summarize some of the main results here.

We consider an anharmonic (nonsinusoidal) periodic function of time, such as a square wave. Then Fourier's theorem states that it can be represented as a Fourier series

$$f(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} A_n \cos n\omega t + \sum_{n=1}^{\infty} B_n \sin n\omega t \tag{2.32}$$

where

$$A_n = \frac{2}{T}\int_0^T f(t)\cos n\,\omega t\,\mathrm{d}t$$
$$B_n = \frac{2}{T}\int_0^T f(t)\sin n\,\omega t\,\mathrm{d}t$$

The symmetry or lack thereof of the function to be analyzed can lead to important simplifications. For example, suppose that the origin has been chosen so that the square wave in question has odd symmetry. Since sine waves have odd symmetry ($\sin t = -\sin(-t)$) and cosine waves even ($\cos t = \cos(-t)$), the

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Fourier series of this square wave can have only sine terms. After only three terms, the general shape of the square wave is reproduced, but clearly it will take many terms (in principle an infinite number) to reproduce the vertical front.

2.1.7 Nonperiodic Waves: Fourier Integral

The previous results on Fourier analysis (synthesis) can be extended from periodic functions to nonperiodic functions (e.g., single pulses) by a simple artifice. If we extend the period T in Equation 2.32 to $T \to \infty$, then we effectively have a single pulse or more generally a transient disturbance $f(t)$ that we can describe by a simple generalization of the series

$$f(t) = \frac{1}{\pi} \int_0^\infty A(\omega)\cos\omega t\,\mathrm{d}\omega + \int_0^\infty B(\omega)\sin\omega t\,\mathrm{d}\omega \tag{2.33}$$

where

$$A(\omega) = \int_{-\infty}^{\infty} f(t)\cos\omega t\,\mathrm{d}t$$
$$B(\omega) = \int_{-\infty}^{\infty} f(t)\sin\omega t\,\mathrm{d}t$$

As an example for a square pulse (see Figure 2.5),

$$\begin{aligned} f(t) &= E_0 \qquad |t| < \frac{T}{2} \\ &= 0 \qquad |t| > \frac{T}{2} \end{aligned} \tag{2.34}$$

which is an even function, the sine term is zero, and

$$\begin{aligned} A(\omega) &= E_0 T \frac{\sin\dfrac{\omega T}{2}}{\dfrac{\omega T}{2}} \\ &= \operatorname{sinc}\frac{\omega T}{2} \end{aligned} \tag{2.35}$$

which is also shown in Figure 2.5. This is a very familiar result in optics when variables t and ω are replaced by x and k. It corresponds to diffraction by a single slit.

It is more economical and standard practice to rewrite Equation 2.33 in complex notation to obtain a Fourier transform pair

$$\begin{aligned} f(t) &= \int_{-\infty}^{\infty} g(\omega)\mathrm{e}^{j\omega t}\mathrm{d}\omega \\ g(\omega) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} f(t)\mathrm{e}^{-j\omega t}\mathrm{d}t \end{aligned} \tag{2.36}$$

where the negative frequency, by Euler's theorem, is nothing more than a way to write the complex conjugate

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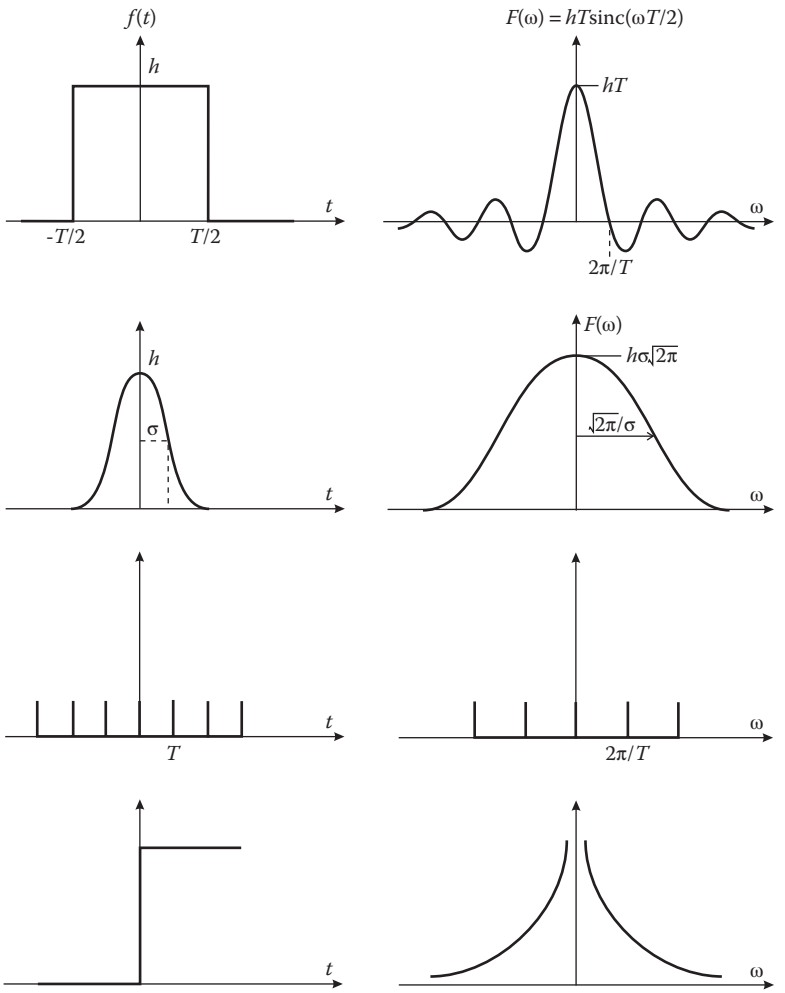

FIGURE 2.5 Some common Fourier transform pairs.

$$e^{j(\pm\omega t)} = \cos\omega t \pm j\sin\omega t \tag{2.37}$$

It is readily seen that dimensionally the members of the Fourier transform pair are the inverse of each other. Moreover, if the pulse is very narrow in t space, it is very wide in ω space and vice versa. Two important examples are the slit function, already shown as having a sine Fourier transform, and the Gaussian, both shown in Figure 2.5. The Gaussian transform can easily be changed into another Gaussian. As a limiting case, consider the Dirac delta function

$$\begin{gathered}\delta(t) = 0 \quad \text{for} \quad t \neq 0 \\ \int_{-\infty}^{\infty} \delta(t)\,\mathrm{d}t = 1\end{gathered} \tag{2.38}$$

which is an infinite spike of unit area at $t = 0$. Then the Fourier transform

$$g(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta(t)e^{-j\omega t}\mathrm{d}t = \frac{1}{2\pi} \tag{2.39}$$

is a constant, independent of frequency.

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The δ function results are a direct demonstration of the bandwidth theorem, which states that

$$\omega \quad t \sim 1 \tag{2.40}$$

Applied to a single pulse, the theorem states that the narrower the pulse, the wider the associated frequency spectrum and vice versa, consistent with the results for the Fourier transform of the Gaussian. We return to the bandwidth theorem in the next section to generalize it to the case of waves and wave packets.

2.2 WAVE MOTION

Waves are universal, presenting themselves in different guises in nature, and they are ubiquitous in the physics and engineering laboratory. They are in fact so common in different areas of science (acoustics, optics, electromagnetics, etc.) that wave motion is usually taught as a subject in its own right in elementary physics courses. What follows is not a substitute for these elementary treatments but rather a summary that enables us to collect the main results in one place, establish notation, and emphasize certain concepts that are important for this book, such as phase and group velocity.

The first question is: What is a wave? In fact, simple intuitive answers to this question can be reformulated in precise mathematical language to provide a test for a given function to decide if it corresponds to wave propagation or not. For the moment, we avoid pathological problems such as strongly scattering, highly dispersive media, and so on, and concentrate on the linear regime in simple, nondispersive media. In this spirit, we then define a wave "as the self-sustaining propagation at constant velocity of a disturbance without change of shape." We can represent the shape of the disturbance by the function $f(x, t)$, a Gaussian $f(x, 0)$ at $t = 0$. The pulse is then propagated at constant velocity V, and at time t, we can describe the same profile in a moving reference frame x' as $f(x')$. Since $x' = x - Vt$ by inspection and there has been no change in shape, we have $f(x, t) = f(x - Vt)$ for any time t. This form $f(x - Vt)$ is characteristic of a wave traveling to the right or in the forward direction. It is easy to see, for the same coordinate system, that a wave propagating to the left would be described by $f(x + Vt)$. This simple rule has a functionality that will become clear throughout the book. For example, according to it, $\sin(\omega t - kx)$ is indeed a wave and $\sin\omega t$ is not; in fact, the latter is clearly an example of harmonic motion of a fixed oscillator, as discussed in Section 2.1.

As for the case of simple harmonic motion for a mechanical oscillator, we determine the equation of motion of the mechanical system under study by combining Hooke's law with Newton's equation of motion. One of the simplest possible examples is that of the transverse vibrations of a string or a cord (see Figure 2.6). For simplicity, we consider the string to be under a certain tension T and to be infinite in length. While the tension T is constant along the string, this is not true for the y component, due to the curvature of the string.

From Figure 2.6, for an element $\mathrm{d}x$

$$\mathrm{d}F_y = (T\sin\theta)_{x+\mathrm{d}x} - (T\sin\theta)_x \tag{2.41}$$

Doing a Taylor's expansion for F_y

$$F(x + \mathrm{d}x) = F(x) + \frac{\partial F}{\partial x}\,\mathrm{d}x \tag{2.42}$$

we have

$$\mathrm{d}F_y = \quad (T\sin\theta)_x + \frac{\partial(T\sin\theta)}{\partial x}\mathrm{d}x + \cdots \quad - (T\sin\theta)_x$$

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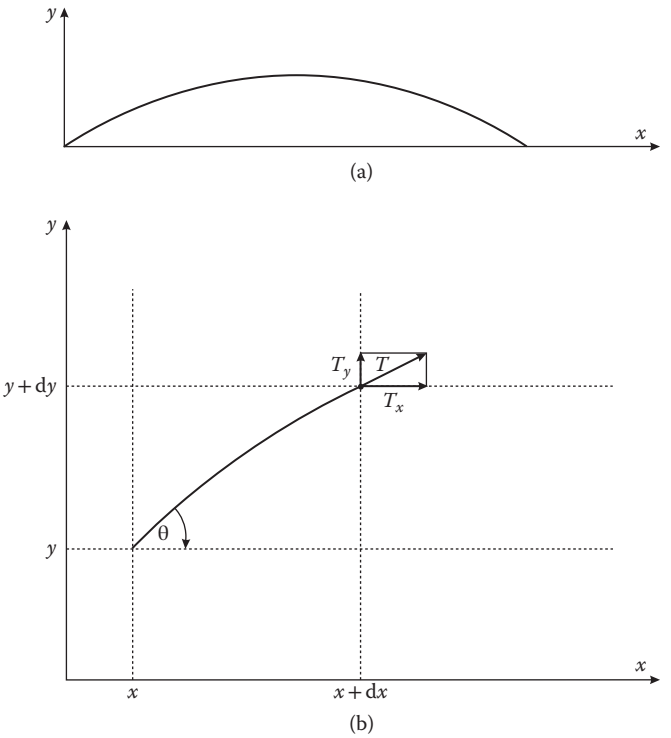

FIGURE 2.6 (a) Vibrating string with fixed end points. (b) Forces on a string element.

so that

$$dF_y = \frac{\partial(T\sin\theta)}{\partial x}dx \tag{2.43}$$

For small displacement (θ) of the string, $\sin\theta \sim \frac{\partial y}{\partial x}$, so that

$$dF_y = T\frac{\partial^2 y}{\partial x^2}dx \tag{2.44}$$

From Newton's law for a string of mass per unit length ρ_l

$$dF_y = \rho_l dx\frac{\partial^2 y}{\partial t^2} \tag{2.45}$$

Combining Equations 2.44 and 2.45, we have the one-dimensional wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{V_0^2}\frac{\partial^2 y}{\partial t^2} \tag{2.46}$$

where

$$V_0^2 = \frac{T}{\rho_l} \tag{2.47}$$

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The form of the wave equation, Equation 2.46, is in fact completely general for all types of waves and the form of V_0^2 is typical for that sort of mechanical system. The tension T that can be applied is proportional to the mechanical stiffness of the system, and this fact can be used to obtain a priori estimates of the sound velocity in a given system. For example, for a given value of ρ_l, the mechanical wave (sound) velocity in a steel cord is going to be much higher than that in a cord made of cooked spaghetti.

2.2.1 Harmonic Waves

For a general wave motion, we write $\psi(x, t) = f(x, t)$, so that the wave equation for ψ is

$$\frac{\partial^2\psi}{\partial x^2} = \frac{1}{V_0^2}\frac{\partial^2\psi}{\partial t^2}3 \tag{2.48}$$

with a general solution of the form

$$\psi = C_1 f(x - V_0 t) + C_2 g(x + V_0 t) \tag{2.49}$$

To summarize the basic wave parameters, we consider a wave profile

$$\psi = A \sin kx \tag{2.50}$$

and if this is propagating to the right, then from the earlier equation

$$\psi = A \sin k(x - V_0 t + \phi) \tag{2.51}$$

and the well-known wave parameters are the following:

- Initial phase angle ϕ
- Wavelength $\lambda = 2\pi/k$
- Wave number k
- Period $T = 1/f$
- Frequency $f = \omega/2\pi$

All of these lead to

$$\omega = V_0 k \quad \text{or} \quad V_0 = \lambda f \tag{2.52}$$

and

$$\psi = A \sin(kx - \omega t + \phi) \tag{2.53}$$

Let us look in more detail at the velocity. We define the phase of the wave as the argument of the harmonic function

$$\equiv kx - \omega t + \phi \tag{2.54}$$

Then the phase velocity is defined as the velocity of propagation of constant phase, for example, that of a wave crest.

Then,

$$= kx - \omega t + \phi = \text{constant} \tag{2.55}$$

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Hence,

$$k\,\mathrm{d}x - \omega\,\mathrm{d}t = 0 \quad \text{for} \quad = \text{constant}$$

so

$$V_{\mathrm{P}} \equiv \left.\frac{\mathrm{d}x}{\mathrm{d}t}\right|_{\phi} = \frac{\omega}{k} \tag{2.56}$$

Alternatively, this result can be obtained using the chain law for partial derivatives from thermodynamics

$$\left.\frac{\partial x}{\partial t}\right|_{\phi} = \frac{-\left.\dfrac{\partial}{\partial t}\right|_{x}}{\left.\dfrac{\partial}{\partial x}\right|_{t}} = \frac{\omega}{k} \tag{2.57}$$

Finally, it is common practice to describe wave motion using the complex exponential. As outlined previously,

$$\psi(x,t) = \mathrm{Re}[A \exp i(\omega t - kx + \phi)] = A\cos(\omega t - kx + \phi) \tag{2.58}$$

In physics, it is common to use the above notation, for example,

$$\psi(x,t) = A \exp i(kx - \omega t)$$

and in engineering it is more common to use the complex conjugate

$$\psi(x,t) = A \exp j(\omega t - kx)$$

where both i and j represent $\sqrt{-1}$.

Both notations are encountered frequently in the literature. For uniformity, we arbitrarily adopt the form $\exp j(\omega t - kx)$ in the rest of the book.

2.2.2 Plane Waves in Three Dimensions

We adopt a three-dimensional coordinate system (x, y, z) with propagation in the direction of the propagation vector $\vec{k} = (k_x, k_y, k_z)$. The wavefront is the locus of points of constant phase at a given time t, so for plane waves it can be represented as a series of parallel planes. If $\vec{r}$ is a position vector from the origin to a point on the wavefront at time t, then the equation of the wavefront is

$$\vec{k} \cdot \vec{r} = \text{constant} \tag{2.59}$$

and so in complex notation we describe the plane wave by

$$\psi(x,t) = A \exp j(\omega t - \vec{k} \cdot \vec{r} + \phi) \tag{2.60}$$

By a simple generalization of the one-dimensional case, we can show directly from the above that, as before,

$$V_{\mathrm{P}} = \frac{\omega}{k} \tag{2.61}$$

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The solution for propagation in an arbitrary direction $\vec{k}$ can be written as

$$\psi(x, y, z, t) = A \exp j(\omega t - [k_x x + k_y y + k_z z]) \tag{2.62}$$

or in terms of direction cosines n_x, n_y, n_z for $\vec{k}$

$$\psi(x, y, z, t) = A \exp j(\omega t - k[n_x x + n_y y + n_z z]) \tag{2.63}$$

where

$$k^2 = k_x^2 + k_y^2 + k_z^2 \tag{2.64}$$

and

$$n_x^2 + n_y^2 + n_z^2 = 1 \tag{2.65}$$

The plane wave equation in three dimensions is usually written in terms of the Laplacian

$$^2 = \frac{\partial}{\partial x^2} + \frac{\partial}{\partial y^2} + \frac{\partial}{\partial z^2} \tag{2.66}$$

so that

$$^2\psi = \frac{1}{V_0^2}\frac{\partial^2 \psi}{\partial t^2} \tag{2.67}$$

2.2.3 Dispersion, Group Velocity, and Wave Packets

Up to now, we have considered the simplest possible model for wave propagation: isotropic, homogeneous, linear, and dispersionless. Some of these simplifications will be removed later, but dispersion is appropriate to consider now. Dispersion basically means that the phase velocity varies with frequency. In optics, dispersion manifests itself in the splitting of white light into its spectral components by a prism or a raindrop. In that case, the dispersion is due to the frequency-dependent movement of the atomic mass. In acoustics, the same effects happen at very high frequencies or with thermal phonons near the Brillouin zone boundaries and the resulting dispersion curves can be measured directly by neutron scattering (Figure 2.7). However, in acoustics, the relevant length scale is 10^5 times larger than in optics, so that for the relatively low ultrasonic frequency range, the wavelengths are quite large, of the order of 100 μm to 1 mm. This is of the same order of magnitude as the critical dimension of the films, plates, wires, and so on, used to guide ultrasonic waves, so we can expect to encounter dispersion in such structures on purely geometrical grounds. Hence, it is essential that we appreciate the consequences of dispersion right from the beginning.

For waves of all types, no information whatsoever is transmitted by the "pure" sinusoidal carrier wave, apart from its characteristic frequency. To transmit information, we need to modulate the carrier with other frequencies, and it is appropriate to consider the velocity of propagation of this modulation, and thus, more generally, the velocity of propagation of information and of energy. The simplest case to consider is that of the wave packet, treated in detail in all standard texts on waves. If several neighboring frequencies are linearly superimposed, they form a wave packet with finite extension in space and a corresponding finite Fourier frequency spectrum. The modulation is somewhat

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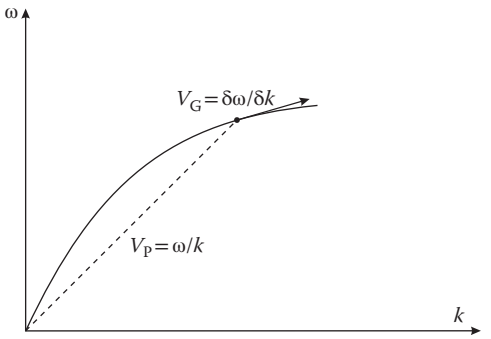

FIGURE 2.7 Typical dispersion curve showing phase velocity and group velocity for one point on the curve.

analogous to the beats for simple harmonic motion considered earlier. The modulation travels at the velocity of this wave packet. For a simple model of two waves with a small difference in frequency

$$\begin{aligned} \psi_1 &= \cos(\omega_1 t - k_1 x) \\ \psi_2 &= \cos(\omega_2 t - k_2 x) \end{aligned} \tag{2.68}$$

the superposition of ψ_1 and ψ_2 gives

$$\psi = 2\cos\ \frac{(\omega_1 + \omega_2)}{2} t - \frac{(k_1 + k_2)}{2} x\ \ \cos\ \frac{(\omega_1 - \omega_2)}{2} t - \frac{(k_1 - k_2)}{2} x \tag{2.69}$$

A whole wave packet can be built up by the superposition of such pairs with a center frequency $\omega_0 = (\omega_1 + \omega_2)/2$ and a modulation frequency $\omega_m = (\omega_1 - \omega_2)/2$. For this simple example, the modulation has velocity $(\omega_1 - \omega_2)/(k_1 - k_2)$. For $\omega_1 \to \omega_2 \to \omega_0$, this goes to $V_G = \partial\omega/\partial k$, the group velocity. Two different forms of V_G for calculation purposes are

$$V_G = \frac{\partial\omega}{\partial k} = V_P + k\frac{dV_P}{dk} \tag{2.70}$$

and

$$\frac{1}{V_G} = \frac{\partial k}{\partial\omega} = \frac{1}{V_P} - \frac{\omega}{V_P^2}\frac{dV_P}{d\omega} \tag{2.71}$$

The bandwidth theorem of simple harmonic motion can be generalized to waves by considering conjugate variables x, k in addition to ω, t. Thus,

$$\omega\ \ t \sim 1 \qquad x\ \ k \sim 1 \tag{2.72}$$

where the latter relation becomes evident from Figure 2.5. The bandwidth relation has its most famous application to wave packets in quantum mechanics where $p = h/\lambda = \hbar k$ is the particle momentum and $E = \hbar\omega$ the particle energy. Thus, we have

$$x\ \ p \sim \hbar \quad \text{and} \quad E\ \ t \sim \hbar \tag{2.73}$$

the celebrated Heisenberg uncertainty principle.

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SUMMARY

Simple harmonic motion refers to harmonic vibrations at a frequency f of a point mass about an equilibrium point. The movement is in general maintained by an external force and is damped by frictional forces. The motion is governed by Newton's second law and Hooke's law, giving a natural frequency $\omega^2 = k/m$.

Phasor is a representation of the vibration in complex plane. The phasor rotates at the phase angle $\theta = \omega t$ and the radius vector is the amplitude of vibration. Two or more phasors can be added algebraically in the complex plane.

Resonance occurs when the imaginary part of the impedance is zero. The resonance can be described by Q or the quality factor, which is a measure of the sharpness of the resonance. There are five different ways of expressing Q, each with a different but complementary physical interpretation.

Fourier series is a way of representing a periodic function as a sum of sines or cosines with argument of an integral multiple of the fundamental frequency. The sine series is used for functions of odd symmetry, the cosine series for even functions.

Fourier integral is a generalization of Fourier series to the representation of pulses by a frequency spectrum. A Fourier transform pair links Fourier representations of a pulse in the time and frequency domain or quantities in spatial and wave number space.

Traveling waves correspond to the self-sustaining propagation of a disturbance in space at constant velocity without change of shape. A progressive or traveling wave is characterized by a functional form $f(kx - Vt)$.

Harmonic wave is a wave at a single frequency ω, described by a sine or cosine function, or in complex notation by $\exp j(\omega t - kx)$.

Phase velocity $V_P = \omega/k$ is the velocity of a wavefront of constant phase.

Group velocity $V_G = \delta\omega/\delta k$ is the velocity of propagation of a wave packet. For lossless or low loss media, it is also the velocity of propagation of energy.

Dispersion describes a situation in which the phase velocity varies with frequency; it occurs in dispersive media.

QUESTIONS

1. Draw a diagram to show how to add two phasors graphically to determine their total amplitude and phase angle. Determine analytical expressions for the latter.
2. Make a graph of the displacement and velocity for a forced simple harmonic oscillator as a function of frequency. Draw the corresponding phasor diagram. Compare the results for oscillators where $R \to 0$ and $R \to \infty$.
3. Consider a triangular waveform as a function of time. Define the amplitude and period. Choose an origin and sketch the first three Fourier components. Comment on the use of sine or cosine functions.
4. Draw two limiting cases (width going to zero or infinity) for the Fourier transform of a Gaussian pulse.
5. Draw the vector diagram corresponding to $\tan\theta$ for simple harmonic motion.
6. Decide which of the following are traveling waves and calculate the appropriate phase velocity:
   i. $f(x, t) = (ax - bt)^2$
   ii. $f(x, t) = (ax + bt + c)^2$
   iii. $f(x, t) = 1/(ax^2 + b)$ a, b, and c are positive constants.
7. Consider a harmonic wave with given ω and k. Give V_P, T, and λ in terms of these quantities.
8. Consider the dispersion curve $w(k) = A\ |\sin ka|$. Plot $w(k)$ over the range $-\pi/a \le k \le \pi/a$. Make plots of $V_P(k)$ and $V_G(k)$. Do likewise for $V_P(\omega)$ and $V_G(\omega)$.

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9. Plot Equation 2.69 for the case where $\omega_1 \gg \omega_2$. Comment on the pertinence of this case for communications.
10. Calculate the group velocity for the following cases where the phase velocity is known:
    i. Transverse elastic wave in a rod

$$V_P = A/\lambda$$

    ii. Deep water waves

$$V_P = A\sqrt{\lambda}$$

    iii. Surface waves in a liquid

$$V_P = A/\sqrt{\lambda}$$

    iv. Electromagnetic waves in the ionosphere

$$V_P = \sqrt{c^2 + A^2\lambda^2}$$

where c is the velocity of light.

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